# Paul Parisot

Software Engineer - Core C++ & Critical Systems

Github: paulogarithm — paul.parisot@epitech.eu

## About

Rigorous software engineer specializing in C/C++ systems and high-performance computing for critical environments.
Strong background in applied mathematics, geometric algorithms, and hardware-accelerated optimization (SIMD).
Committed to high software quality standards, strict requirement traceability, and comprehensive automated testing.

## Skills

**Technical Expertise**: C, C++ (Modern standards up to C++20), Assembly, Go, Linux Systems
**Math & Computing**: Geometric Algorithms, 3D Linear Algebra (4x4 Matrices), SIMD Vectorization
**Quality & Testing**: Automated Testing, CI/CD Pipelines, Test Coverage Optimization, Technical Documentation
**Workflow & Tools**: Agile/Scrum, Jira (Requirement Tracking), Git, Version Control, Strict Process Compliance

## Professional Experience

**Faurecia / Forvia - Software Engineer (Embedded C++ / Industrial Quality)**

- Developed mission-critical low-level C++ network modules for real-time embedded communications.
- Implemented complex navigation and path-generation algorithms under strict industrial software quality processes.
- Tracked and managed technical requirements using **Jira** within a highly structured Agile/Scrum framework.
- Produced rigorous technical documentation ensuring end-to-end traceability of software evolutions.

**Gno.land - Junior Software Engineer (Distributed Systems & Testing)**

- Maintained and optimized open-source runtime core modules, focusing on deterministic execution and code correctness.
- Mitigated **6 critical runtime vulnerabilities**, contributing over **10k lines of code** backed by thorough testing.
- Developed extensive regression and automated integration test suites under zero-tolerance CI/CD pipelines.

**Epitech / IONIS - Technical Assistant & Quality Reviewer**

- Mentored and evaluated over **400 students** on systems programming (C, C++, Assembly) and algorithmic precision.
- Enforced strict coding standards, mandatory automated testing paradigms, and thorough memory-leak audits.
- Authored technical reference documentation and complex evaluation specifications for junior cohorts.

## Technical Projects & Applied Mathematics

**rodox – Low-Latency Systems & Compute Engine**

Designed a modular dynamic engine architecture in C/C++ natively for Linux using cache-friendly layouts.
Implemented real-time 3D spatial coordinate layers using **4x4 transformation matrices** and linear algebra.
Optimized high-frequency computational loops through hardware-accelerated **SIMD vectorization** pipelines.
**Skills:** C++, Linux OS, 3D Linear Algebra, SIMD Optimization, Memory Management, Modular Architecture

**Raytracer – 3D Geometric Renderer from Scratch**

Developed a high-precision software rendering engine in C++ focused on mathematical and spatial modeling.
Coded vector calculus operations, matrix inversions, and intensive **ray-object intersection and tracking algorithms**.
**Skills:** C++, Coordinate Systems, Geometry Algorithms, Spatial Tracking, Object-Oriented Design

## Theoretical Research & Systems Engineering

- **Advanced Computer Science**: Deep study of deterministic systems, formal logic, and complexity theory.
- **Runtime Assembly Injection (cclass)**: Engineered a low-level C mechanism utilizing Linux `mmap` pages to dynamically patch compiled assembly stubs, proving absolute control over binary execution layers.
- **Project Management**: Acquired formal methodologies in team alignment and risk assessment at McGill University.

## Education

**Epitech, Paris** 2022 – 2027
Master's Degree in Information Technology (Master's level diploma)
Level-7 Certification - Expert in Software Engineering
Specialization: Systems Architecture

**McGill University, Montréal** 2025 – 2026
Certificate in Management
Technical Leadership & Strategy